Leihan Wang

📫 leihanwa@buffalo.edu | 🏠 leihanwang.github.io | 🎓 Leihan Wang - Google Scholar

EDUCATION

School of Engineering and Applied Sciences
University at Buffalo, SUNY (Buffalo, New York, The United States)
Bachelor of Science in **Computer Science** August 2024 - Present

College of Artificial Intelligence and Automation
Hohai University (Nanjing, Jiangsu, China)
Bachelor of Engineer in **Artificial Intelligence** August 2022 - July 2024

I am an undergraduate student transferring from *Hohai University* to *University at Buffalo* 😊

RESEARCH EXPERIENCE

University at Buffalo, SUNY (Buffalo, New York, The United States) August 2026 – Present

Research Student, hosted by Prof. Yaxiong Xie

- Conducted systems research on **mobile SoC architectures for on-device LLM inference**, focusing on heterogeneous CPU/GPU/NPU execution, Qualcomm Hexagon/HVX vector processing, memory hierarchy, and data movement across mobile accelerators.
- Investigated modern **NPU-based LLM inference systems including NPUGen and ShadowNPU**, analyzing architectural techniques such as KV-cache recomputation, sparse attention, accelerator offloading, and the limitations imposed by NPU programmability and operator support.
- Explored **architecture-aware optimization opportunities for mobile LLM inference** by connecting prefill, decoding, and attention workloads with hardware characteristics including SIMD parallelism, cache/VTCM behavior, memory bandwidth, instruction scheduling, and power/thermal constraints.

Clemson University (Clemson, South Carolina, The United States) May 2026 - August 2026

Remote Research Student, hosted by Prof. Xiaoyong (Brian) Yuan

- Developed a multimodal foundation-model alignment framework for drug discovery, integrating molecular, protein, textual, and biomedical representations.
- Built fusion strategies to align heterogeneous foundation models and improve molecule–target association, drug property prediction, and biomedical reasoning.
- Incorporated domain-specific chemical and biological priors to enhance cross-modal representation learning and foundation-model-based drug discovery.

The University of Georgia (Athens, Georgia, The United States) December 2025 – February 2026

Remote Research Student, hosted by Prof. Zhenlin An

- Developing a PE-AV model and beamforming-based audio-visual perception system for AR glasses.
- Building an AR audio-visualization framework that localizes sound sources and visually maps audio signals to corresponding sound-emitting objects in the environment.
- Incorporating visual priors to improving sound source association, spatial audio understanding, and object-level audio-visual reasoning.

University at Buffalo, SUNY (Buffalo, New York, The United States) February 2025 - Present

Research Student, hosted by Prof. Haipeng Cai

- Developed an enhanced BODE-GEN framework for LLM-driven program synthesis, using Bayesian Optimization to discover task-adaptive prompts and exemplars from a subset of HumanEval+ problems.
- Integrated an auxiliary LLM with a Bayesian-optimized prompt memory to select task-specific prompt–exemplar combinations for unseen coding problems, improving generalization beyond a single globally optimized prompt.
- Reduced reliance on expensive per-task Bayesian Optimization by optimizing only 20 HumanEval+ tasks offline and transferring the learned prompt memory to the remaining 144 tasks for single-generation pass@1 evaluation.

Hohai University (Nanjing, Jiangsu, China) September 2024 – February 2025

Research assistant, hosted by Prof. Zaipeng Xie

- Developed **FedSTAR and FedHAN**, federated learning frameworks addressing **spatiotemporal/data heterogeneity, model heterogeneity, and label noise** across decentralized trajectory prediction and heterogeneous client models. FedSTAR specifically combines alignment, drift control, and performance-aware aggregation. FedHAN focuses on reliability under heterogeneous architectures and noisy labels.
- Designed robust local and global optimization mechanisms, including **two-stage contrastive alignment, adaptive model-drift regularization, reliability-weighted aggregation, consistency-based sample weighting, and prototype-based classifier refinement**, improving stability and knowledge sharing across heterogeneous clients.
- Evaluated the frameworks across **ETH/UCY and MNIST/CIFAR-10/CIFAR-100**, with FedSTAR achieving the lowest average ADE/FDE among federated baselines under heterogeneous trajectory partitions and maintaining performance under dynamic client participation, while FedHAN improved accuracy by up to **5.76% on CIFAR-10 and 4.67% on CIFAR-100** under heavy label noise.

PROJECT EXPERIENCE

Huaqing Water Treatment Company (Jilin, Jilin, China) June 2023 - August 2023

Project Engineer

- Developed an **automated water-treatment control system on Siemens S7-300 PLCs using STEP 7**, using real-time flow, pressure, pH, and process measurements to govern filtration and multi-pump start/stop logic.
- Implemented **multi-pump rotation with a dedicated standby pump** and feedback-based chemical dosing that adjusts dosing-pump output with water level and process conditions to maintain target treatment conditions and reduce unnecessary equipment loading.
- Built **fault detection, alarm, and automatic failover logic** for pump or sensor abnormalities, switching to the backup pump to support stable unattended operation; designed the project for a **KingView 7.5 SP2 SCADA** environment.

PUBLICATIONS

[2] **(IEEE BigData 2025)** Zaipeng Xie, Zishu Zhou, Xuanyao Jie, Xing Gao, Han Xu, and **Leihan Wang, "**FedHAN: Robust Federated Learning under Model Heterogeneity and Label Noise." Accepted to *2025 IEEE International Conference on Big Data* (IEEE BigData 2025), 2025.

[1] **(EAI CollaborateCom 2025)** Zaipeng Xie, Dingxu Sun, Peixin Li, Ziyang Ye, and **Leihan Wang, "**FedSTAR: A Federated Learning Framework for Reliable Trajectory Prediction under Spatiotemporal Heterogeneity." Accepted to *21st EAI International Conference on Collaborative Computing: Networking, Applications and Worksharing* (EAI CollaborateCom), 2025.

SELECTED COURCES GRADES

- **Algorithms and Complexity 98/100** **Artificial Intelligence 91/100**
- **Algorithms and Data Structures 96/100** **Introduction to UAV 96/100**